PUSHFORWARD OF A QUASI-COHERENT MODULE ALONG A QUASI-COMPACT QUASI-SEPARATED MORPHISM OF ALGEBRAIC SPACES IS QUASI-COHERENT

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-morphisms.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof uses equation (equation-representable-pushforward) of the same chapter as a labelled formula; the formula itself is consumed inside the proof span, so no extra context span was added.

Editorial note. The eight references of the counted spans are recorded as non-load-bearing navigation pointers; the argument that consumes them stays inside the counted proof span.

Editorial note. Preamble: the three native excerpt spans of sources/stacks-spaces-morphisms/preamble.tex (lines 8-13, 14-18 and 154-217). The chapter file itself carries no preamble text.

Allowed standard prerequisites (editorial).

- Quasi-coherence of a module on an algebraic space is tested on affine schemes etale over it. Properties of Spaces, Lemma spaces-properties-lemma-characterize-quasi-coherent, used at proof lines 2134-2138. Reduces the claim to a statement about the restriction of f_*F to an affine etale V over Y.
- The base change formula for pushforwards along an etale V -> Y. equation-representable-pushforward, used at proof lines 2144-2147. Identifies the restriction with the pushforward of the restriction along the base change f'.
- The exact sequence computing the pushforward from an etale presentation. Lemma lemma-compute-pushforward, used at proof lines 2155-2159. Presents f_*F as the kernel of a map between the pushforwards along the two projections of R.
- Quasi-coherence of pushforwards of quasi-coherent modules under quasi-compact quasi-separated morphisms of schemes. Descent, Proposition descent-proposition-equivalence-quasi-coherent-functorial, used at proof lines 2164-2168. Makes a_*(F|_U) and b_*(F|_R) quasi-coherent.
- A kernel of a map of quasi-coherent modules is quasi-coherent. Properties of Spaces, Lemma spaces-properties-lemma-properties-quasi-coherent, used at proof lines 2169-2172. Concludes for f_*F.

Source reference. Original proposition 03LC: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4708–4708. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original equation 04CF: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 5689–5689. Navigation pointer into the Stacks

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Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 03HF: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4565–4565. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 03KL: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4681–4681. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 03M8: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4740–4740. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 03M0: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4732–4732. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 03M1: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4733–4733. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 03H6: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4555–4555. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Lemma 0.2. *Let S be a scheme. Let $f : X \to Y$ be a morphism of algebraic spaces over S. If f is quasi-compact and quasi-separated, then f_* transforms quasi-coherent $\mathcal{O}_X$-modules into quasi-coherent $\mathcal{O}_Y$-modules.*

Proof. Let $\mathcal{F}$ be a quasi-coherent sheaf on X. We have to show that $f_*\mathcal{F}$ is a quasi-coherent sheaf on Y. For this it suffices to show that for any affine scheme V and étale morphism $V \to Y$ the restriction of $f_*\mathcal{F}$ to V is quasi-coherent, see Properties of Spaces, Lemma 03M0. Let $f' : V \times_Y X \to V$ be the base change of f by $V \to Y$. Note that f' is also quasi-compact and quasi-separated, see Lemmas 03HF and 03KL. By (04CF) we know that the restriction of $f_*\mathcal{F}$ to V is f'_* of the restriction of $\mathcal{F}$ to $V \times_Y X$. Hence we may replace f by f', and assume that Y is an affine scheme.

Assume Y is an affine scheme. Since f is quasi-compact we see that X is quasi-compact. Thus we may choose an affine scheme U and a surjective étale morphism $U \to X$, see Properties of Spaces, Lemma 03H6. By Lemma 03M8 we get an exact sequence

$$0 \to f_*\mathcal{F} \to a_*(\mathcal{F}|_U) \to b_*(\mathcal{F}|_R).$$

where $R = U \times_X U$. As $X \to Y$ is quasi-separated we see that $R \to U \times_Y U$ is a quasi-compact monomorphism. This implies that R is a quasi-compact separated scheme (as U and Y are affine at this point). Hence $a : U \to Y$ and $b : R \to Y$ are quasi-compact and quasi-separated morphisms of schemes. Thus by Descent, Proposition 03LC the sheaves $a_*(\mathcal{F}|_U)$ and $b_*(\mathcal{F}|_R)$ are quasi-coherent (see also

the discussion preceding this lemma). This implies that $f_*\mathcal{F}$ is a kernel of quasi-coherent modules, and hence itself quasi-coherent, see Properties of Spaces, Lemma 03M1. □